MILITARY SPECIFICATION
OBTURATING BANDS, PLASTIC, RIGID, HIGH IMPACT, POLYVINYL CHLORIDE (MOLDED)

1. SCOPE

1.1 This is the applicable specification for Rigid, High Impact, Polyvinyl Chloride Plastic Obturating Bands (Molded), for artillery ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION
Federal
L-P-406 - Plastics, Organic, General Specifications, Test Methods

STANDARDS
Federal
FED-STD-595 - Colors
Military
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking of Shipments

3. REQUIREMENTS

3.1 General.- All parts shall comply with the requirements of the applicable drawings, referenced specifications and the following:

3.2 Material.- The material shall consist of polyvinyl chloride resin of high molecular weight, suitably compounded with heat and light stabilizers, a suitable elastomer, lubricants and pigments. No re-grind or scrap will be allowed.

3.3 Molded parts.- Parts made in accordance with the applicable drawings shall be produced by the injection molding method. All tests required by this document shall be performed after the parts have been maintained at 73° F for a minimum of 24 hours.

3.3.1 Workmanship.- The molded parts shall be uniform in color, free of cracks, burn marks, flashing and contamination. Surface irregularities resulting from cutting off sprues and gates shall be removed by grinding and buffing.

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3.4 Breaking load.- The minimum breaking load shall be as specified on the applicable drawing.

3.5 Heat stability.- The molded part shall not char, brown or blacken when subjected to 400° ± 2° F, for a period of 30 minutes.

3.6 Acetone resistance.- The molded part shall not flake or delaminate when submerged in acetone for a period of 30 minutes, except within 1/2-inch of cut edges.

3.7 Color.- The molded part shall be a light gray, approximately that of Color Number 36270 of FED-STD-595.

3.8 Hidden defects.- When parts which have been inspected, tested and accepted, contain inherent defects subsequently exposed, the contractor may be required to replace the defective material without additional expense to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 Lot size.- Under this specification, a lot shall be considered to be any quantity of bands procured under one contract but shall be limited to 5000 or less. If the procurement under one contract is greater than 5000 bands, then multiple lots shall be employed, each lot being 5000 or less.

4.3 Inspection provisions.

4.3.1 Molded parts.- Provisions for dimensional and visual inspections are given in the applicable specification for the projectile assembly.

4.3.2 Breaking load.- Ten molded parts per lot shall be subjected to this test but not less than 10 per 5000 parts produced shall be tested. Failure of two or more parts to comply with the requirement of 3.4 shall be cause for rejection of the lot. If one part fails then an additional 10 parts shall be tested. Failure of one or more of these 10 parts to comply with the requirements shall be cause for rejection of the lot.

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4.3.3 Heat stability.- Ten molded parts per lot shall be subjected to this test, but not less than 10 per 5000 parts produced shall be tested. Failure of two or more parts to comply with the requirement of 3.5 shall be cause for rejection of the lot. If one part fails then an additional 10 parts shall be tested. Failure of one or more of these 10 parts to comply with the requirements shall be cause for rejection of the lot.

4.3.4 Acetone resistance.- Ten molded parts per lot shall be subjected to this test, but not less than 10 per 5000 parts produced shall be tested. Failure of two or more parts to comply with the requirement of 3.5 shall be cause for rejection of the lot. If one part fails then an additional 10 parts shall be tested. Failure of one or more of these 10 parts to comply with the requirements shall be cause for rejection of the lot.

4.4 Test methods and procedures.

4.4.1 Molded parts.- Reference shall be made to the applicable projectile assembly specification for dimensional and visual inspection.

4.4.2 Breaking load.- A fixture shall be employed which consists of two semicircular steel discs with a diameter slightly less than that of the band to be tested and with a thickness of (at least) that of the band. The configuration of the outer diameter of the discs shall conform to that of the inner diameter of the band to be tested. The fixture is placed together to form a round disc and the band is then fitted around the fixture. The discs shall be attached to the self-aligning grips of a tensile testing machine by means of two suitable harnesses (see figure I). The band shall be positioned such that the weld line is at right angles to the direction of pull. With the band and fixtures in place, the tensile testing machine is operated at a constant rate of 2.0 inches per minute until the band breaks. The amount of force required to break the band will be recorded as the breaking load.

4.4.3 Heat stability.- The bands shall be placed in a circulating air oven or cabinet pre-heated to 400° ± 2° F. After a period of 30 minutes, the bands shall be removed and examined for evidence of charring, browning or blackening of the surface.

4.4.4 Acetone resistance.- One specimen 3 ± 1/16 inches long, shall be taken from each band. The section of the band from which the specimen is taken will not include a gate or weld area. The specimen shall be submerged in acetone for a period of 30 minutes. The specimen shall then be examined for evidence of flaking or delamination.

5. PREPARATION FOR DELIVERY

5.1 General.- The packaging, packing and marking requirements specified herein, apply only to direct purchases by or direct shipments to the Government.

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5.2 Packing.- Unless otherwise specified, all materials shall be packed for domestic shipment. Shipping containers shall be uniform in size, and shall be designed to inclose the contents in a snug, tight-fitting manner.

5.2.1 Domestic packing.- Unless otherwise specified, the material shall be packed in substantial commercial containers constructed to insure acceptance by common or other carrier for safe transportation (at the lowest rate) to the point of delivery.

5.3 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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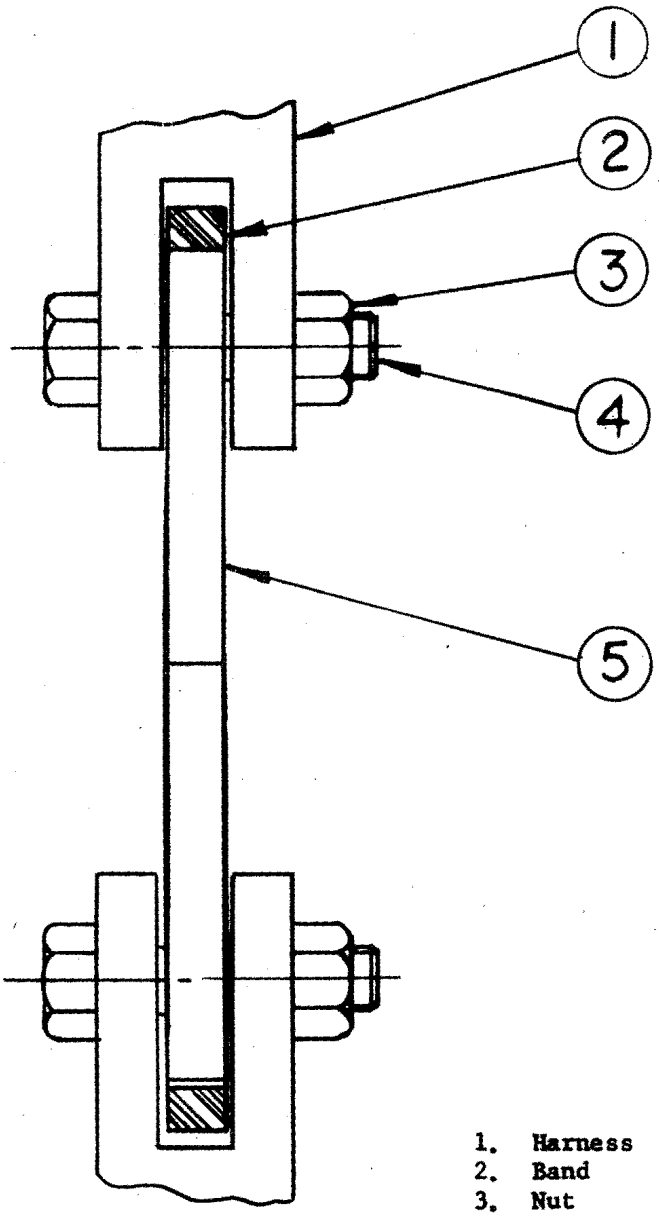
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1. Harness
2. Band
3. Nut
4. Bolt
5. Disc

FIGURE 1

Fixture for Breaking Load Test

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